Dr. Chilperic Armel Foko Kuate

Computational Scientist | Multiscale Modeller
Scientific Software Developer | Applied Mathematics
Mechanistic models • numerical simulation • inference and optimisation • scientific software • reproducible evidence
Düsseldorf, Germany +49 176 55 20 83 05 chilpericarmel@gmail.com
github.com/chilperic chilperic.github.io
ORCID 0000-0002-0140-7588 Scholar profile

Profile

- Computational scientist and multiscale modeller with a PhD in Quantitative and Theoretical Biology and formal training in applied mathematics; research spans cell populations, metabolic networks, and coupled plant–environment physiology.
- Build interpretable mechanistic models and numerical workflows using ODE/DAE and stochastic systems, nonlinear dynamics, parameter estimation, optimisation, sensitivity, uncertainty, and identifiability analysis.
- Develop reproducible scientific software, including FokoLab, a browser-native modelling platform designed around user-defined models, simulation, analysis, diagnostics, visualisation, and exportable evidence.

Multiscale modelling and computational science

Across biological scales	Enzyme kinetics and metabolic networks; cellular population dynamics; organ/leaf biochemistry, heat balance and hydraulics; environmental forcing and adaptive phenotypes.
Applied mathematics	Nonlinear dynamical systems, stability and bifurcation reasoning, stochastic processes, branching processes, inverse problems, constrained optimisation, and numerical analysis.
Inference & validation	Parameter fitting, identifiability, Sobol/Morris sensitivity, uncertainty analysis, simulation ensembles, numerical reference checks, and explicit limits of interpretation.
Scientific software	Reproducible Python and browser-native modelling tools with editable inputs, diagnostics, visualisation, testing, documentation, and data/report export.

Core methods and tools

Modelling	ODE/DAE systems, kinetic rate laws, energy and hydraulic balances, nonlinear dynamics, stochastic processes, branching processes, master equations.
Numerics	NumPy/SciPy, CVODE/SUNDIALS, CasADi/IPOPT, Pyomo, lmfit, CMA-ES, parameter fitting, constrained optimisation.
Analysis	Sobol/Morris sensitivity, identifiability, uncertainty analysis, simulation ensembles, stability analysis, PCA/statistical diagnostics.
Programming	Python, Julia, R, MATLAB; JavaScript/HTML/CSS for browser-native scientific software.
Workflow	Git/GitHub, Jupyter, Linux, HPC exposure, testing, LaTeX, reproducible pipelines, technical documentation.

Research and modelling experience

Nov 2025–Present

Independent Computational Scientist & Scientific Software Developer

Düsseldorf, Germany

- Architect and develop FokoLab, a browser-native scientific modelling platform spanning deterministic and stochastic dynamics, optimisation, sensitivity, population/evolution, statistics/ML, symbolic analysis, reproducibility and validation diagnostics.
- Refactor research code into documented, testable repositories with explicit model assumptions, numerical checks, reusable examples, and reproducible computational workflows.
- Advance two first-author manuscripts on multiscale modelling of plant adaptation across the C3–C4 continuum while completing a professional German B2 programme.

Nov 2022–Oct 2025

Postdoctoral Researcher

Germany

CEPLAS / Institute of Computational Cell Biology, HHU Düsseldorf

- Developed a multiscale thermo-hydraulic-biochemical framework coupling C3/C4 carbon assimilation, mesophyll/bundle-sheath heat balance, stomatal conductance, water transport, soil state, irradiance, humidity, and temperature-dependent enzyme kinetics.
- Combined Sobol sensitivity, simulation ensembles, and CMA-ES optimisation across contrasting climates to identify influential traits, physiological trade-offs, and environment-specific adaptive strategies.
- Worked with experimental collaborators and supervised 2 MSc and 2 BSc projects, translating biological questions into quantitative models and interpretable computational analyses.

Research and modelling experience — continued

Marie Skłodowska-Curie PhD Researcher — Aug 2019–Aug 2022

Institute of Quantitative and Theoretical Biology, HHU Düsseldorf — Germany

- Developed two ODE-based models of hepatic fatty-acid metabolism and de novo synthesis, connecting qualitative dynamical-systems analysis with quantitative simulation of experimental time courses.
- Derived conditions for metabolic bistability using polynomial/root analysis and stability theory; fitted semi-mechanistic enzyme-kinetic models and assessed robustness with sensitivity analysis.
- Contributed enzyme-kinetic formalism to a cross-institutional differentiable constraint-based modelling framework; work published in *Bioscience Reports* and *Metabolic Engineering*.

Research Fellow — Nov 2017–Mar 2019

AIMS Cameroon and AIMS Rwanda — Cameroon / Rwanda

- Developed stochastic and deterministic models of T-cell proliferation using branching-process, master-equation, delay and compartmental formulations.
- Built likelihood-based fitting workflows for CFSE-type data and extracted interpretable cell-population parameters from experimental distributions.
- Delivered 200+ teaching hours in mathematical modelling, ODEs, stochastic processes, algebra, and Python.

Selected projects

FokoLab Open platform — Browser-native scientific modelling platform for user-defined models, simulation, fitting, sensitivity, optimisation, population/evolution, diagnostics, visualisation and reproducible export.

Photosynthesis climate-adaptation model GitHub — Python/CasADi mechanistic model coupling C3/C4 biochemistry, heat balance, hydraulics, soil state, climate forcing, optimisation, and sensitivity workflows.

Fatty-acid de novo synthesis model GitHub — Kinetic ODE model with enzyme-rate laws, parameter estimation, validation, and sensitivity analysis.

Fatty-acid metabolism bistability GitHub — Nonlinear metabolic model exploring switching behaviour, steady states, and robustness in lipid metabolism.

Master work: T-cell proliferation models GitHub — AIMS Ghana MSc thesis, *Reverse-engineering of T-cell proliferation dynamics*: Smith–Martin DDE, Cyton model, branching-process thinking, CFSE-type fitting, and population-response inference.

Education

2019–2023	PhD, Quantitative and Theoretical Biology, Heinrich Heine University Düsseldorf. Dissertation: dynamic computational modelling of fatty-acid synthesis and metabolism; defended March 2023. Supervisor: Prof. Dr. Oliver Ebenhöh.
2017–2019	Master 2, Mathematics and Applications, University of Yaoundé I. Advanced numerical analysis, dynamical systems, inverse problems.
2015–2016	MSc, Mathematical Sciences, AIMS Ghana. Thesis: *Reverse-engineering of T-cell proliferation dynamics*; Smith–Martin DDE, Cyton model, CFSE-type data, and population-dynamics inference.
2009–2012	BSc, Mathematics, University of Douala. Algebra, analysis, geometry.

Selected publications

Dourado, V.H., Foko Kuate, C.A., Liebermeister, W., Lercher, M.J. Growth mechanics and the emergence of metabolic oscillations in growing cells. *bioRxiv* preprint, 2025. DOI: 10.1101/2025.06.24.661369. Second author; contributed mathematical framework for metabolic-oscillation modelling.

Foko Kuate, C.A., Ebenhöh, O., Bakker, B.M., Raguin, A. Kinetic data for modeling the dynamics of the enzymes involved in animal fatty-acid synthesis. *Bioscience Reports*, 43(7), BSR20222496, 2023. DOI: 10.1042/BSR20222496.

Wilken, S.E., Besançon, M., Kratochvíl, M., Foko Kuate, C.A., Trefois, C., Gu, W., Ebenhöh, O. Interrogating the effect of enzyme kinetics on metabolism using differentiable constraint-based models. *Metabolic Engineering*, 74, 72–82, 2022. DOI: 10.1016/j.ymben.2022.09.002.

Two first-author manuscripts in preparation on mechanistic modelling of plant adaptation across the C3–C4 continuum.

Training, leadership, languages, awards

Training	Bayer B4U University Mentoring Programme, completed 2025 (mentor: Benjamin Buer); PoLiMeR Marie Curie ITN doctoral training; Advanced AI and Data Science Workshop, HeiCAD; Hands-on Course on Optimal Control with CasADi.
Leadership	C4 Plant Project Group Leader; Early Career Researchers' Representative, Marie Curie ITN; supervised 2 MSc and 2 BSc projects.
Languages	French: native; English: fluent; German: B1+ (currently completing B2 Berufssprachkurs; exam expected Nov 2026).
Awards	Marie Skłodowska-Curie ITN Fellowship (PoLiMeR); Post-AIMS Research Bursary; F.K.A. Allotey Meritorious Award, AIMS Ghana.